Intelligence, Cost and Adaptation: Technology in the War in Ukraine

Patrick J. Lowry*

Abstract

Russia's war against Ukraine has tested many assumptions about how a large conventional war in Europe would be fought. In this paper I use open sources to examine what the war reveals about the performance of modern military technologies, including those used to collect intelligence. I make three arguments. First, the commercialisation and socialisation of open-source intelligence have produced an *intelligence-as-a-service* model, in which commercial providers, civil-society analysts and ordinary citizens supplement state collection, at some legal and ethical cost. Second, unable to achieve technological overmatch, Ukraine has pursued an asymmetric strategy of cost imposition built on large numbers of inexpensive uncrewed systems and affordable long-range strike, although Russian precision fires show that high-end capabilities retain their value. Third, the war has rewarded a federated, innovation-led industrial base that turns battlefield feedback into new capability within months rather than years. Together, these findings suggest that agility, information advantage and cost-effective adaptation now matter as much as mass and platform sophistication.

Keywords: Ukraine; Russia; open-source intelligence; commercial intelligence; uncrewed systems; cost imposition; defence-industrial base; military innovation

1. Introduction

In this paper I examine Russia's war against Ukraine and what it reveals about the performance of modern military technologies, including those used to collect intelligence. In particular, I examine how the war has overturned assumptions held by think tanks and governments about how a modern conflict would unfold (Rickli and Mantellassi, 2024), and how Ukraine has adapted to reshape the tactical battlefield, allowing a weaker state to hold its own against a larger one. I draw only on open sources.

2. A war that confounded expectations

By most conventional measures, Ukraine entered the war with a weaker deterrence and defence posture than Russia. Yet the heavier losses have fallen on the aggressor. An analysis by the Center for Strategic and International Studies, published in January 2026, estimated around 1.2 million Russian casualties against 500,000 to 600,000 Ukrainian (Ward, 2026). Russia's own expectations had been very different: captured planning documents indicate that it expected to defeat Ukraine within ten days and to annex the country by August 2022 (Zabrodskyi et al., 2022). More than four years on, governments, think tanks, civil society and defence companies have had to revise their early assessments of a war that has settled into attrition, in which success is measured in losses inflicted rather than ground taken. The war has undermined conventional

*The views expressed in this paper are the author's own. They do not represent the policy or internal thinking of His Majesty's Government or the Ministry of Defence, and the paper draws only on published sources.

metrics of military strength across land, sea and air, and has become a costly campaign in which a nuclear-armed state spends far more than it gains.

Against this background, I argue that three developments define how technology has performed in this war, and where advantage now lies within an ongoing conflict.

1. **A new intelligence-as-a-service model.** The commercialisation of intelligence and the growth of open-source intelligence (OSINT) have multiplied the data streams available to decision-makers. Forward-leaning partners, commercial providers and civil society have created a decentralised network of intelligence nodes that improves situational awareness. The resulting volume of data can match some aspects of more advanced intelligence systems, and the participation of private companies and citizens gives collection a more social character.

2. **An asymmetric strategy of imposing cost.** Because Ukraine cannot achieve technological overmatch against Russia, it has not followed traditional procurement models built on expensive, high-end capabilities. Instead, it has pursued cost imposition through large numbers of inexpensive uncrewed systems and affordable long-range strike. This form of precise mass threatens far more costly Russian assets, alters Russia's decision-making calculus, and is now being adopted by other militaries, Russia's among them.

3. **A redefined industrial base.** Advanced technology no longer guarantees success below the nuclear threshold. To keep pace, Ukraine has built a new industrial base that favours innovation, pace, mobility and economies of scale. Its short procurement loop, driven by feedback and intelligence from the front, can outpace the multi-year cycles of far more costly high-end programmes.

These developments help explain why a campaign intended to be decisive became a prolonged financial and military quagmire. Both sides have had to learn, adapt, sustain wartime economies and develop new tactics, and the war has exposed the limits of pre-war assumptions about the role of modern capabilities and intelligence in a European conflict. Western planners should nonetheless be cautious about generalising from it. Some Russian practices reflect a particular context, and in some domains Russia has shown restraint.

3. Intelligence as a service

Intelligence is central to a state's ability to deter and defend. Its purpose is to support decision-making, so that countermeasures and actions yield strategic benefit, and it is often associated with a state's ability to shorten its observe–orient–decide–act cycle, John Boyd's "OODA loop" (Boyd, 2018). For Ukraine, intelligence has been a key enabler, valued not only for its operational benefit but also as a measure of allied resolve. On the battlefield, however, its effectiveness is limited by three factors: speed, dissemination and quality.

Before the war, states spent billions to make their intelligence bespoke, tailored and able to penetrate deep into an adversary's decision-making. That effort concentrated on refining the intelligence cycle of collection, processing, analysis and dissemination, a cycle that had not been tested in a high-intensity war in Europe. Ukraine has combined the bespoke intelligence provided by its allies with the sheer volume of OSINT, and the integration of commercial technologies has irrevocably altered the cycle. Meng (2025) cites a US Defense Intelligence Agency estimate that around 80 per cent of intelligence now originates from open sources, and describes a modern cycle that draws on cross-platform data streams, including official documents, social-media content and commercial satellite imagery. The breadth of these streams, illustrated in figure 1, means that a sufficient intelligence picture can now be built from open information, and built at pace.

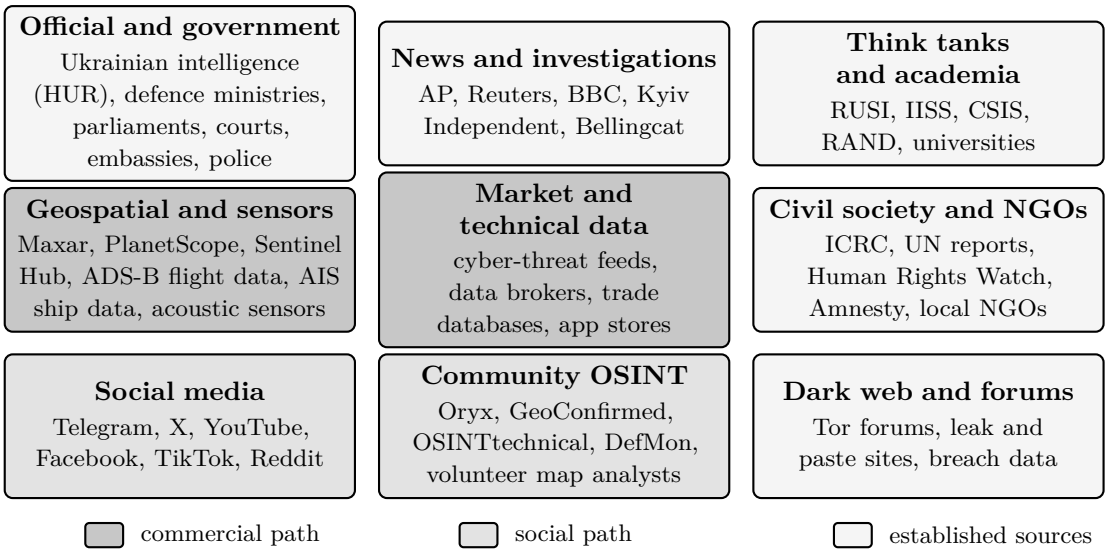


Figure 1. The English-language OSINT source landscape relevant to the war, grouped by category with examples. Adapted from Meng (2025); shading has been added to distinguish the commercial and social paths discussed in § 3 from established sources.

This enhanced role of OSINT follows two distinct paths: commercialisation and socialisation. In the first, companies supplement, and in places replace, state capabilities. Firms such as Palantir, Planet Labs, BlackSky and Clearview AI have created, and put into practice, a new "intelligence as a service" (Nagl and Hackett, 2026). Ukraine gains faster and better intelligence, while the companies gain the first live test of their products in a high-intensity war since 1945. Ukraine has welcomed them in its battle for information advantage. The US Army War College's study of the war's second year highlights Ukraine's use of unclassified information gathered through such companies: because it could largely be released or shared, it helped Ukraine gain information advantage, integrate with allies, build tactical and operational information networks, and engage domestic and foreign audiences (Nagl and Hackett, 2026).

The commercialisation and partial public availability of intelligence have also created an ecosystem in which civilian organisations act as external validators. Bellingcat, for example, uses open-source tools, commercial imagery and social-media tracking to verify troop movements, identify Russian equipment and debunk false narratives (Huminski, 2023). This semi-transparent check on official intelligence builds confidence in conflict analysis. It has also contributed to what Ford and Hoskins (2022) call "participative war", in which civilians contribute analysis that is useful to military analysts and helps to counter disinformation.

The Ukrainian government has gone further, actively encouraging civilians to act as intelligence nodes by using smartphone applications to warn of air raids and to report Russian troop movements and positions (Phythian and Strachan-Morris, 2024). This decentralisation creates an unprecedented crowdsourced sensor network, but it also raises profound legal and ethical questions. A civilian who reports enemy positions risks being regarded as taking a direct part in hostilities, and so losing the protection from attack that international humanitarian law gives to civilians (Melzer, 2009). In occupied areas, the same activity exposes civilians to reprisal if it is discovered. Participatory intelligence therefore shifts risk onto the population it is meant to protect.

Together, this mix of new and established tactics, techniques and procedures allows Ukraine to draw on whichever source performs best for a given task. It has enriched decision-making and made Ukraine's intelligence more resilient to the loss of any one provider. When the United States paused intelligence sharing in March 2025, and Maxar suspended Ukraine's access to its

imagery for about a week (Oliinyk, 2025), Ukrainian units fell back on their own drones, the Delta situational-awareness system and a crowdfunded radar satellite (Yermak and York, 2025).

4. An asymmetric strategy of imposing cost

Across the modern military-industrial complex, states have prioritised technological advantage, treating scientific and technological strength as a key indicator of success. This has driven a continual cycle of demand for ever more premium, bespoke and costly systems, in the expectation that they would consistently outmatch an adversary's capabilities. This has historically been a Western approach, and Western sanctions have sought to deny Russia the components it needs for advanced systems. Russia has responded by accepting lower standards in order to sustain output, producing weapons that are cheaper and more numerous but less reliable, on the judgement that mass matters in an attritional war (Boulègue, 2025).

Ukraine recognises that it cannot achieve conventional or technological overmatch against Russia. It has therefore taken an asymmetric approach to imposing cost on Russia's military advantage. Primarily through affordable technology, above all an expanded role for uncrewed vehicles in every domain (UXVs) (Singh and Singh, 2025), Ukraine is building its own form of lower-cost mass: the rapid, localised production of expendable uncrewed systems that can degrade Russian forces continuously without relying on expensive, bespoke Western platforms.

The availability, rapid configurability, open-source design, ease of use and flexibility of cheap commercial off-the-shelf drones have changed the economics of combat. A first-person-view (FPV) drone costing from a few hundred to a few thousand US dollars can destroy a tank worth around $3 million, a task that previously called for a Javelin missile costing about $100,000. By comparison, Russia's Lancet loitering munition costs an estimated $35,000 to $60,000, and one counter-drone interceptor costs about $150,000 while carrying a payload comparable to that of a $300 kamikaze drone (Molloy, 2024). Electronic warfare became the main defence against cheap drones, and the drones adapted in turn. Fibre-optic control lines, which leave no radio link to jam, have made drones effectively unjammable at ranges of more than 30 kilometres, and have become critical kit for both sides (Sutea, 2026).

In a complex battlefield, however, high-end capabilities such as Russian precision fires have kept their value. While much of the fighting has become a contest of cheap drones, Russia's cruise, ballistic and hypersonic missiles have delivered significant strategic effect and forced Ukraine to expend scarce and costly air-defence interceptors. Improved Iskander ballistic missiles have proved increasingly hard to stop: between 1 and 24 October 2025, Ukraine intercepted only 14 of 78, or 17 per cent, with a limited force of six Patriot batteries (Cranny-Evans and Kaushal, 2025).

Ukraine cannot realistically field enough high-end interceptors to deny Russia's most advanced systems, and its dependence on US-made munitions leaves it short of them. Its answer has been to impose costs of its own through affordable long-range strike. The FP-5 Flamingo cruise missile, taken by the Ukrainian company Fire Point from concept to production in about nine months, is reported to cost up to €1 million, against €1.5 million to €4 million for a US Tomahawk, while offering a longer range of up to 3,000 kilometres and a heavier warhead (Yan, 2025).

Unable to match Russian mass or to rely entirely on premium Western weapons, Ukraine has pioneered what might be called an "affordable bespoke" procurement model. By combining cheap commercial drones with domestic precision strike, it imposes an unfavourable cost ratio on its adversary, and shows that rapid technological iteration and economic sustainability matter as much as traditional firepower.

5. A redefined industrial base

Russia's illegal war has shown the importance of adaptability. A tight OODA loop provides insight, cheap precise mass delivers fires at scale, and bespoke capabilities maximise the cost imposed on an adversary. All of this, however, depends on an industrial base that is robust, innovation-driven and resilient.

Out of necessity, Ukraine has become a leader in agile procurement. Russian strikes have badly damaged its legacy industrial base, and new capacity has had to be built (Dee et al., 2025). In its place, Ukraine has developed a federated base of small and medium-sized producers that provides resilience, redundancy and protection for a skilled workforce. It has also forged a direct link between the front line and production: the front offers a live environment in which to test new concepts, and commanders' feedback flows directly into the development of systems. This has increased both the scale of production and the range of products.

Ukraine also benefits from access to Western technology that can be integrated into its products. Programmes such as the UK–Ukraine TechExchange connect Ukrainian defence start-ups with British firms and investors, and Western companies increasingly regard Ukraine as the best available testing ground (Hill, 2024). Leonardo, for example, plans to test components of its Michelangelo air-defence system in Ukraine before NATO trials begin (Defense Express, 2026). An agile, federated and increasingly connected industrial base therefore serves two purposes. It refines how Ukraine fights and adapts its capability, from drones to domestic cruise missiles, and it binds Western industry, and with it Western governments, more closely to Ukraine.

By establishing a federated, innovation-driven defence ecosystem, Ukraine has built a system in which battlefield intelligence, operational feedback and industrial production adapt continuously. The model allows rapid iteration of technologies, from uncrewed systems to long-range strike, so that Ukraine can respond quickly to changing conditions on the battlefield and integrate Western technology as it arrives. It offers a new model of defence innovation, one that sustains technological advantage through quick feedback, real-world testing, experimentation and federated manufacturing, even against a stronger adversary.

6. Conclusion

The war in Ukraine challenges traditional metrics of military power. It shows that agility and continuous adaptation can offset a substantial part of an opponent's conventional mass. By drawing on a decentralised intelligence-as-a-service ecosystem, using uncrewed systems to impose disproportionate costs on expensive legacy assets, and integrating battlefield feedback rapidly into a reconfigured defence-industrial base, Ukraine has reshaped how this war is fought.

These lessons should be drawn with care. Russian precision fires show that high-end capabilities still matter, attrition still decides much of the outcome, and participatory intelligence carries real legal and ethical costs. The conflict nonetheless suggests that future warfare will be shaped as much by information advantage, cost-effective technological integration and the speed of innovation as by sheer military bulk.

Acknowledgements. This paper develops a formative essay prepared for a postgraduate certificate at University College London.

Use of AI tools. Claude was used to edit and typeset this paper. The author is responsible for its content.

References

Mathieu Boulègue. Russia's struggle to modernize its military industry: How sanctions, war and 'innovation stagnation' are weakening Moscow's capabilities. Research paper, Chatham House, London, July 2025. URL https://www.chathamhouse.org/2025/07/russias-struggle-modernize-its-military-industry.

John R. Boyd. *A Discourse on Winning and Losing*. Air University Press, Maxwell Air Force Base, AL, 2018. Edited by Grant T. Hammond.

Sam Cranny-Evans and Sidharth Kaushal. Iskander: an improved Russian missile tests Ukraine's air defence. RUSI Commentary, Royal United Services Institute, 18 November 2025. URL https://www.rusi.org/explore-our-research/publications/commentary/iskander-improved-russian-missile-tests-ukraines-air-defence.

Stuart Dee, K. Fedina, K. Suman-Chauhan, E. Graham, D. Hill, and A. Gibson. Looking ahead: Enablers of innovation and scale for the future of Ukraine's defence-industrial base. Research Report RR-A3833-6, RAND Europe, Cambridge, 2025.

Defense Express. Ukraine to test Leonardo's new Michelangelo Dome air defense before NATO trials. Defense Express, 12 March 2026. URL https://en.defence-ua.com/weapon_and_tech/ukraine_to_test_leonardos_new_michelangelo_dome_air_defense_before_nato_trials-17808.html.

Matthew Ford and Andrew Hoskins. *Radical War: Data, Attention and Control in the Twenty-First Century*. Hurst, London, 2022.

John Hill. Ukraine defence industries are set to integrate with the West. Army Technology, 2 April 2024. URL https://www.army-technology.com/newsletters/ukraine-defence-industries-are-set-to-integrate-with-the-west.

Joshua C. Huminski. Russia, Ukraine, and the future use of strategic intelligence. *PRISM*, 10(3): 8–25, 2023. URL https://www.jstor.org/stable/48743420.

Nils Melzer. *Interpretive Guidance on the Notion of Direct Participation in Hostilities under International Humanitarian Law*. International Committee of the Red Cross, Geneva, 2009.

Wei Meng. The intelligent evolution of open-source intelligence: focusing on the International Legion of Defence Intelligence of Ukraine. Preprints.org, preprint (not peer reviewed), 6 October 2025. URL https://doi.org/10.20944/preprints202510.0414.v1.

Oleksandra Molloy. Drones in modern warfare: Lessons learnt from the war in Ukraine. Australian Army Occasional Paper 29, Australian Army Research Centre, Canberra, 2024. URL https://doi.org/10.61451/267513.

John A. Nagl and Michael T. Hackett. *A Long, Hard Year: Russia–Ukraine War Lessons Learned 2023*. US Army War College Press, Carlisle, PA, 2026.

Vira Oliinyk. Maxar brings back access to satellite imagery for Ukrainians after a week-long pause. AIN.UA, 12 March 2025. URL https://en.ain.ua/2025/03/12/maxar-brings-back-access-for-ukraine.

Mark Phythian and David Strachan-Morris. Intelligence & the Russo-Ukrainian war: introduction to the special issue. *Intelligence and National Security*, 39(3):377–385, 2024. doi: 10.1080/02684527.2024.2330132.

Jean-Marc Rickli and Federico Mantellassi. The war in Ukraine: Reality check for emerging technologies and the future of warfare. Geneva Paper 34/24, Geneva Centre for Security Policy, Geneva, April 2024. URL https://gcsp.ch/publications/war-ukraine-reality-check-emerging-technologies-and-future-warfare.

Inderjit Singh and Jashandeep Singh. The emerging role of drones in shaping present and future conflicts. *Zulfaqar Journal of Defence Management, Social Science & Humanities*, 8(2), 2025.

doi: 10.58247/jdmssh-2025-0802-04.

Vlad Sutea. Fiber-optic drones have emerged as critical kit for both Russia and Ukraine. UkraineAlert, Atlantic Council, 24 February 2026. URL https://www.atlanticcouncil.org/blogs/ukrainealert/fiber-optics-drones-have-emerged-as-critical-kit-for-both-russia-and-ukraine/.

E. J. Ward. Millions of casualties, metres gained: Russia's Ukraine war nears two million losses with battlefield gains measured in feet. LBC, 29 January 2026. URL https://www.lbc.co.uk/article/russia-ukraine-war-casualties-two-million-battlefield-gains-metres-5HjdRZF_2/.

Oleksandr Yan. The Economist: Ukrainian Flamingo cruise missile costs less than one million euros. Militarnyi, 29 August 2025. URL https://militarnyi.com/en/news/the-economist-ukrainian-flamingo-cruise-missile-costs-less-than-one-million-euros/.

Natalia Yermak and Chris York. 'We will adapt' — Ukraine's soldiers say after US intel cut. The Kyiv Independent, 7 March 2025. URL https://kyivindependent.com/we-will-adapt-ukraines-soldiers-say-after-us-intel-cut.

Mykhaylo Zabrodskyi, Jack Watling, Oleksandr V. Danylyuk, and Nick Reynolds. Preliminary lessons in conventional warfighting from Russia's invasion of Ukraine: February–July 2022. Special report, Royal United Services Institute, London, 30 November 2022.